

Practice Questions

Test your comprehension by answering the following questions:

1. What is the head of an array or string?
2. What is the tail of an array or string?
3. What are the three questions this chapter presents for each recursive algorithm?
4. What is the leap of faith in recursion?
5. What do you need to understand about the recursive function you are writing before you can take a leap of faith?
6. How does a linear data structure such as an array or string resemble a tree-like structure?
7. Does the recursive `sum()` function involve any backtracking over the data it works on?
8. In the flood fill program, try changing the `im` variable's strings to create a *C* shape that is not fully enclosed. What happens when you attempt to flood-fill the image from the middle of the *C*?
9. Answer the three questions about recursive solutions for each of the recursive algorithms presented in this chapter:
 - a. What is the base case?
 - b. What argument is passed to the recursive function call?
 - c. How does this argument become closer to the base case?

Then re-create the recursive algorithms from this chapter without looking at the original code.